Vasily Dunaev

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Robotics R&D engineer with an MSc in Applied Mathematics and a BSc in Robotics, focused on robot dynamics, control, numerical simulation, and mathematical modelling of physical systems. I work across geometric mechanics, simulation, and model-based and learning-based control, with hands-on experience building robotic systems.

Skills

Robot kinematics and dynamics, model-based control, numerical simulation, reinforcement learning;
Differential geometry, Lie groups and Lie algebras, geometric mechanics, optimisation;
Python, C++, Julia, MATLAB/Simulink, MuJoCo, Git, CAD, LaTeX.

Education

- **University of Twente, MSc in Applied Mathematics** Sep 2023 – Aug 2025
  Enschede, The Netherlands
  Grade: 9.8/10
  Thesis: Geometric structure of Cosserat elasticity [3]. Supervisor: Frederic P. Schuller.
- **University of Twente, Erasmus+ Exchange** Feb 2022 – Feb 2023
  Enschede, The Netherlands
  Grade: 9/10
  Took MSc courses on Robust Control, Optimal Control, Nonlinear Control and Scientific Computing.
- **ITMO University, BSc in Robotics** Sep 2019 – Jun 2023
  St Petersburg, Russia
  Grade: 4.6/5
  Designed and organised an experimental advanced BSc programme in Robotics focused on fundamental subjects. Initially created for my group, the programme was later adopted for subsequent cohorts.

Experience

- **Reinforcement Learning Control Project** May 2026
  Developed a hybrid controller for full SE(3) pose tracking with a torque-controlled UR5e robot simulated in MuJoCo. Combined a geometric spring-damper impedance controller with an RL policy trained with SAC that outputs a virtual end-effector wrench correction. [repo]
- **Graduate Researcher (University of Twente)** Sep 2023 – Sep 2026
  Developed a differential-geometric formulation of the kinematics and dynamics of Cosserat continua, solving the open problem of providing necessary and sufficient global compatibility conditions [1]. In preparation for submission to the Journal of Nonlinear Science, draft on request.
- **Instructor (University of Twente)** Sep 2022 – Nov 2025
  Co-taught a master's course on robot modelling, kinematics, and dynamics with Stefano Stramigioli. Covered rigid transformations, twists, wrenches, robot kinematics, dynamics, and torque control. Delivered lectures and tutorial sessions; designed assignments and exams; produced lecture notes [2] and video lectures. [repo]
- **Research Intern (KdV Institute, University of Amsterdam)** Mar 2025 – Jun 2025
  Built reusable Julia software for structure-preserving numerical integration on matrix Lie groups, including simulation tools for rigid-body dynamics on SO(3) and SE(3), preserving Casimirs to machine precision. [report]
- **Member, later Engineer (Youth Robotics Lab, ITMO University)** Sep 2013 – Feb 2022
  Selected activities:
  - **Robot-Artist**
    Led a team of eight building a painting robot. Implemented its motion control in C++ and Python and its stroke-sequence generation using genetic and simulated-annealing algorithms; designed and built the hardware. [photos and videos]
  - **ELSA**
    Within a team of three, created a humanoid robot that plays the flute. Designed the CAD model and built the robot's 3D-printed hands. RoboCup Junior 2019 winner. [video in Russian, English CC available]
  - **Strawberry Fields**
    Developed an autonomous strawberry-harvesting prototype; trained a CNN for berry detection and designed a soft robotic gripper. Won 1st place at the World Robot Olympiad 2018. [video]
  - **Teaching**
    Developed and taught a course on CAD 3D modelling and 3D printing.

Publications

[1] **V. Dunaev**, F. P. Schuller, and F. Califano, "Geometric structure of Cosserat continua and global compatibility conditions," to be submitted to *Journal of Nonlinear Science*, 2026.

[2] V. Dunaev, *Modelling, Dynamics, Kinematics. Lecture notes*. University of Twente, 2025.

[3] V. Dunaev, "Geometric structure of Cosserat elasticity," MSc thesis, University of Twente, Enschede, Aug. 2025.

[4] A. A. Peregudin, A. M. Popov, E. R. Akhmetgaliev, A. D. Bugrova, M. A. Bystramovich, V. A. Stakanov, and **V. A. Dunaev**, "Virtual laboratory for game-based control systems education," *IFAC-PapersOnLine*, vol. 55, no. 17, pp. 344–349, 2022, 13th IFAC Symposium on Advances in Control Education ACE 2022.

Selected awards

Recipient – **University of Twente Scholarship** 2023 – 2025
Enschede, The Netherlands | Value: €12,000 per year

1st and 2nd Places – **RoboCup Junior** 2019
Sydney, Australia | OnStage category

1st Place – **World Robot Olympiad** 2018
Chiang Mai, Thailand | Open category (Senior)

1st Place – **RobotChallenge** 2018
Beijing, China | Freestyle category

Languages

English – C1, French – B1, Russian – native.